Name: _____ Roll No: _____

Thapar Institute of Engineering & Technology, Patiala
Department of Mathematics

EVESEM-2024-25 (Mid-Semester Test)

Programme: **M.Sc. Mathematics and Computing** Course Code: **PMC212**
Course Name: **Data Structure &Algorithms**
Total Marks: **30**; Duration: **2 Hours** Course Coordinator: **Dr. Farhan Musanna**
Instructions: *All Questions are compulsory. Assume all recurrences are algorithmic. Array indexing is 0-based. Assume missing data suitably.*

1. (a) Solve using Substitution Method:

$$T(n) = 2T\left(\frac{n}{2}\right) + n$$

Hence, solve $T(n) = 2T(\sqrt{n}) + \Theta(\log n)$ [4M, CO1, L1]

(b) Let $a \geq 1$ and $b > 1$ be constants, and let $f(n)$ be a function defined for real numbers $n \geq 1$. Let the asymptotic behavior of the function $g(n) = \sum_{j=0}^{\lfloor \log_b n \rfloor} a^j f\left(\frac{n}{b^j}\right)$ defined for $n \geq 1$. Then show that if there exists a constant $\epsilon > 0$ such that $f(n) = O\left(n^{\log_b a - \epsilon}\right)$, then $g(n) = O\left(n^{\log_b a}\right)$. [3M,CO1, L2]

2. (a) Convert $(A+B)*(C-D/E\hat{}F)-G$ to postfix and hence evaluate the postfix using Stacks.Show all intermediate steps in the stack.

Operator Precedence & Associativity	
$\hat{}$ (exp)	Right-associative (highest)
$/, *$	Left-associative
$+, -$	Left-associative (lowest)

Given $A = 2, B = 3, C = 10, D = 8, E = 2, F = 3, G = 5$ [5M,CO1, L3]

(b) Given an array of integers, the **Next Greater Element (NGE)** for an element x is the first greater element to the right of x. If no such element exists, the NGE is -1. Using Stacks, give a $O(n)$ algorithm to determine the NGE. Example Input :$[4, 5, 2, 25]$, Output: $[5, 25, 25, -1]$ [4M,CO1, L3]

3. (a) Write the recurrence relation for the worst case of Merge Sort. Give the algorithm for the Merge Operation used in Merge Sort for sorting in a non-decreasing order. Is Merge Sort In-Place and is it stable? [4M,CO1, L4]

(b) Give the algorithm/pseudocode for the partition function in QuickSort that works well for sorted data and data with duplicate values. You may take the example $[1, 1, 1, 1, 1]$ and show your working. Is Quick Sort an in-place sorting algorithm? [3M,CO1, L3/4]

4. (a) Given the array [1, 2, 3 , 4 , 5 , 6 , 7 , 8] Convert it into a MAX HEAP using the Sift-Down method. Show the final array representation. (b) What is the time complexity of this process for an input size n? Justify your answer. Starting from the max heap obtained, perform the deletion of the maximum element. Show all intermediate steps. [4M,CO1, L3]

(b) Show the working of an algorithm that sorts the array [2,1,0,0,1,2,0,0,1,2,1] in non-decreasing order in a single pass. What is the name of the algorithm? Is it in-place? [3M,CO1, L3]